Mitosisphere May Challenge

Thank you for taking the 2025 Mitosisphere May Challenge. The exam consisted of 8 questions with a max of 16 points.

Question 1

Lebron James has just retired from the NBA to lock in for USABO. However, he does still miss being the king of the league, and he wonders what made him so good. He can partially attribute his success to his muscular system. Help Lebron identify the true statement about muscles below.

A. Cardiac muscle cells have shorter action potentials compared to skeletal muscle cells, because they need to contract rapidly and rhythmically.

B. Most muscles contain both fast and slow twitch fibers, but muscles of the eye and hand are exclusively fast-twitch.

C. Since glycolysis produces ATP rapidly, all fast-twitch fibers only do glycolysis without aerobic respiration.

D. Tetanic contractions are most common in cardiac muscle cells.

E. Smooth muscle cell contractions happen when calmodulin binds to troponin, removing tropomyosin from myosin-binding sites on the actin thin chains.

Solution: A is incorrect because cardiac muscle cells actually have longer action potentials due to the plateau phase (caused by L-type Ca^{2+} channels), which prevents tetanus and allows rhythmic contractions. B is true and logically makes sense as both of these organs are specialized for rapid movement. C is incorrect; type IIa (fast oxidative fibers) also do some aerobic respiration. D is incorrect because cardiac cells do not undergo tetanus due to the long refractory period. E is incorrect because this describes striated muscles like cardiac and skeletal, not smooth muscle which lacks troponin.

Question 2

Infrqtion decided to test his novel drug, ubiquitinab, on Tanish, a child he locked in his basement. Ubiquitinab has been found to cause increased excretion of bile salts based on prior testing on mice. Assuming that he was a healthy child prior to the administration on Ubiquitinab into his mouth and that any feedback loop mechanisms related to bile salts haven't been activated yet, which of the following molecules would be found to have found at lower levels in Tanish's body?

A. Maltose

B. Cholesterol

C. Arginine

D. Blood plasma K^+

E. $CaCl_2$

Solution: Due to the increased excretion of bile salts, we are looking for something that is found in large quantities in bile. Among the listed options, only cholesterol fits this description. Therefore, the answer is (B) Cholesterol.

Question 3

Cobalamin (vitamin B12) deficiency often results in a form of macrocytic anemia caused by the accumulation of megaloblasts. Which of the following is the most likely mechanism by which this occurs?

A. Conversion of methylcobalamin into 5'-deoxyadenosylcobalamin in environments with low amounts of B12 cause an inability to synthesize histone variants.

B. A lack of cobalamin causes the inhibition of an enzyme that normally plays a role in the displacement of the N-groups in its porphyrin rings.

C. Cobalamin deficiency directly prevents the demethylation of tetrahydrofolate to convert homocysteine into methionine, preventing its usage in other forms.

D. A deficiency in cobalamin, which is normally an allosteric upregulator of dihydrofolate reductase, directly lowers folate's effects on cell division.

E. A decrease in the conversion of hypoxanthine to inositol.

Solution: Vitamin B12 deficiency interferes with the conversion of homocysteine to methionine, a process requiring methylcobalamin. This causes folate to be trapped in a methylated form, impairing DNA synthesis and leading to macrocytic anemia. The correct answer is (C).

Question 4

Nebulin decided to sequence the DNA of the species *I.somerase*. Unfortunately, he is a dumbass and accidentally used the wrong sequencing method that resulted in only nucleosome-bound DNA being sequenced. Which of the following lab techniques did Nebulin accidentally use on *I.somerase*?

A. Coimmunoprecipitation

B. Southern Blotting

C. ChIP

D. Ion Torrent Sequencing

E. Illumina Sequencing

Solution: ChIP (Chromatin Immunoprecipitation) isolates DNA-protein complexes, particularly those involving histones. Thus, the answer is (C) ChIP.

Question 5

Robert is interested in how bacteria evolve over time in order to confer resistance to certain antibiotics. He has two strains of the novel diploid bacterium Tree Coli, 1 strain from a year ago, and 1 strain from the present day. The year old strain is sensitive to both erythromycin and penicillin due to homozygosity at the erm- and pen- genes. The new strain however, is homozygous for the erm+ and pen+genes, and is thus resistant to both. (Homozygosity of both erm+ and pen+ are needed to confer resistance) Robert crosses these two strains, and then self-crosses the F1 generation and plates them on a petri dish with both erythromycin and penicillin. To his surprise though, he notices a 17% survival rate on his dishes. He concludes the genes must be linked, in cM, how far apart are the erm and pen genes?

A. 35 cM

B. 34 cM

C. 18 cM

D. 9 cM

E. 16 cM

Solution: Since homozygosity for both erm^+ and pen^+ is needed for resistance, only offspring with both homozygous dominant genotypes survive.

The F1 generation is heterozygous for both genes. When the F1 self-crosses, the expected proportion of homozygous dominant offspring for both genes if unlinked would be $\frac{1}{4} \times \frac{1}{4} = \frac{1}{16} = 6.25\%$.

However, Robert observes a 17% survival rate, indicating the genes are linked. Using the recombination frequency r*, the probability of non-recombinant gametes is* $\frac{1-r}{2}$*, and recombinant gametes is* $\frac{r}{2}$.

The probability of homozygous resistant offspring is:

$$\left(\frac{1-r}{2}\right)^2 = 0.17,$$

Solving for r*:*

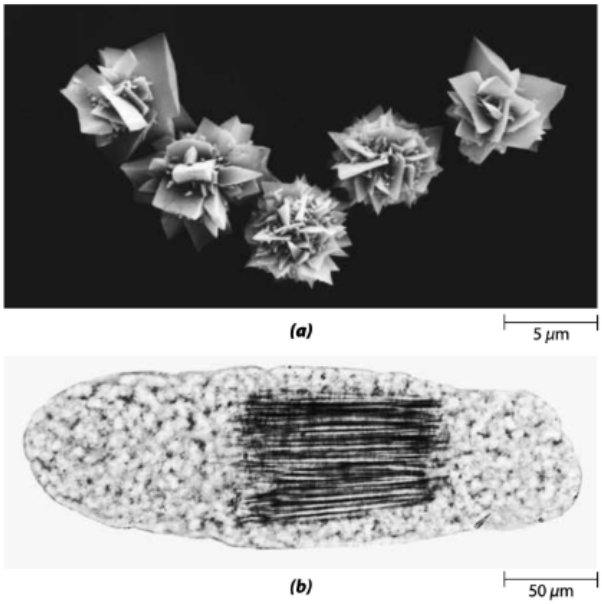
$$\frac{1-r}{2} = \sqrt{0.17} \approx 0.4123,$$

$$1 - r = 0.8246,$$

$$r = 0.1754,$$

Therefore, the distance between the erm and pen genes is approximately $\boxed{18\ cM}$.

Question 6



(i) Identify the structures labeled (a) and (b) in the figure above:

(ii) Determine what the structures are made of:

Solution: Druses are star-like, while raphides are needle-like. Both are made of calcium oxalate and are involved in plant defense.

Question 7

A forest ecologist surveys a small section of woodland and counts four different tree species with the following numbers of individual trees:

Species	Number of Individuals
Species A	10
Species B	30
Species C	60
Species D	100

Solution: Species counts: *A = 10, B = 30, C = 60, D = 100; Total = 200*

$$H = -\sum p_i \ln(p_i)$$

$$= -\left(\frac{10}{200}\ln\frac{10}{200} + \frac{30}{200}\ln\frac{30}{200} + \frac{60}{200}\ln\frac{60}{200} + \frac{100}{200}\ln\frac{100}{200}\right) = 1.14$$

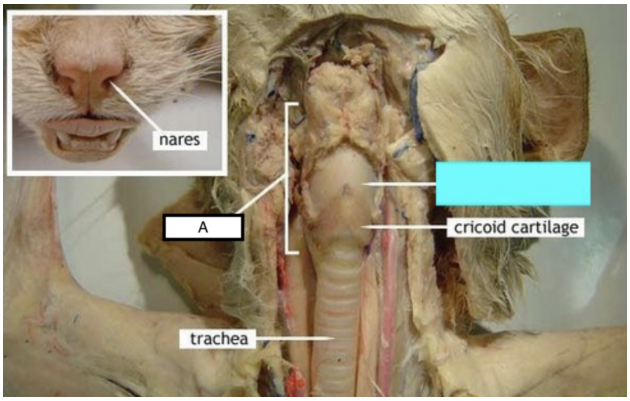
Question 8

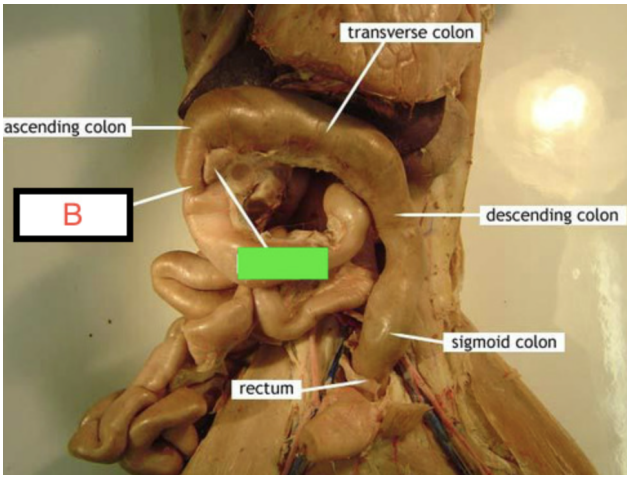
What are the differences between acoelomates, pseudocoelomates and coelomates? Provide an example of each.

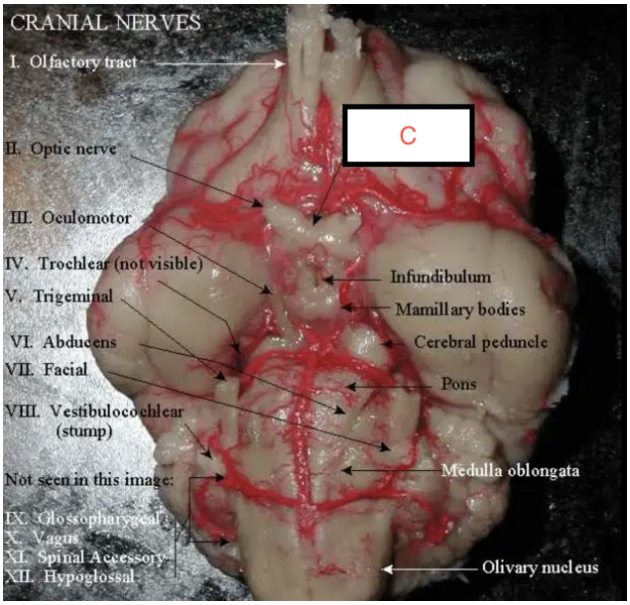
Solution:

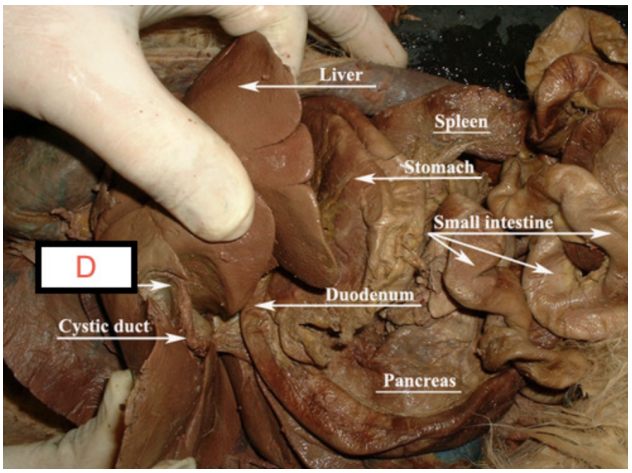
- **A:** *The larynx, located directly before the trachea. It is distinct from the pharynx, which is positioned more posteriorly and internally.*
- **B:** *The cecum, an outpocketing at the beginning of the large intestine.*
- **C:** *The optic chiasm, identifiable as the anterior brain structure where the optic nerves cross over.*
- **D:** *The gallbladder, situated just beneath the liver.*

Histology! You are conducting a cat dissection in your class and your teacher has decided to quiz you on what the following organs are(true story). Be as specific with your descriptions as you can.


nares
A
cricoid cartilage
trachea


transverse colon
ascending colon
B
descending colon
sigmoid colon
rectum


CRANIAL NERVES
I. Olfactory tract
C
II. Optic nerve
III. Oculomotor
IV. Trochlear (not visible)
V. Trigeminal
VI. Abducens
VII. Facial
VIII. Vestibulocochlear (stump)
Not seen in this image:
IX. Glossopharyngeal
X. Vagus
XI. Spinal Accessory
XII. Hypoglossal
Infundibulum
Mamillary bodies
Cerebral peduncle
Pons
Medulla oblongata
Olivary nucleus



Solution:

- ***A:*** *The larynx, located directly before the trachea. It is distinct from the pharynx, which is positioned more posteriorly and internally.*
- ***B:*** *The cecum, an outpocketing at the beginning of the large intestine.*
- ***C:*** *The optic chiasm, identifiable as the anterior brain structure where the optic nerves cross over.*
- ***D:*** *The gallbladder, situated just beneath the liver.*

Tiebreaker!

Name every hormone you can think of and where they are produced.
Solution: This is just a tiebreaker meant to help jog your memory of hormones post USABO season.